



PROJECT DESIGN PHASE



Solution Architecture



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NAME
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PROJECT
OptiCrop – Smart
Agricultural
Production
Optimization Engine



MAX MARKS
4 Marks



Overview

Solution architecture is a process of designing a machine learning based web application that connects a farmer's soil and climate inputs with a crop recommendation. It bridges the gap between crop selection challenges and an intelligent, data-driven software solution.

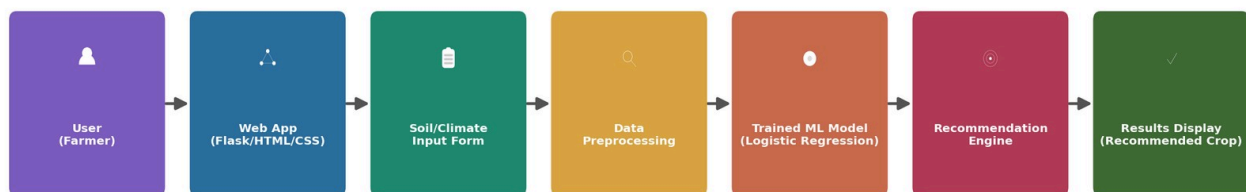


What This Document Covers

- Develops a reliable machine learning solution for recommending crops.
- Describes the system architecture, workflow, and functionality used to generate the crop recommendation.
- Provides a structured framework for developing, deploying, managing, and maintaining the crop recommendation system.



Figure 1 — Architecture and Data Flow of the OptiCrop System



Architecture and data flow of the OptiCrop crop recommendation system



STAGE	COMPONENT	DESCRIPTION
1	User (Farmer)	Enters soil and climate readings through the web interface.
2	Web Application (Frontend)	HTML / CSS / Flask form for soil and climate input.
3	Data Preprocessing	Cleaning, outlier removal, and feature formatting.
4	Trained ML Model	Logistic Regression classifier trained on clustered crop data.
5	Recommendation Engine	Generates the crop prediction from the model output.
6	Results Display	Shows the recommended crop and a short suitability description to the user.